Crypto Lab 1 - Shishir Vuppala

Challenge 5: Never Painted by the Numbers

flag: `cs409{keystr3am_reus3_1n_th3_b19_2026}`

Solution

The vulnerability is that the counter value is predictable and is incremented by a fixed amount after every encryption. Since CTR mode generates a keystream based on the key, nonce, and counter, and the key and nonce stay the same during our session with the server, we can use the predictability of the counter value to get the keystream values that will be used in a future flag query. We can do this by sending a very long plaintext to recover the portion of the keystream corresponding to future counter values.

First, we send a `!flag` request to determine the length of the flag. Then we send a sufficiently long plaintext and XOR the ciphertext received with the known plaintext to get the keystream. We can then send another `!flag` request, and take the keystream starting at offset 480 and XOR it with the flag ciphertext, and therefore, decrypt the flag.

Code

The script used to solve this challenge is written in `solution_template.py`